Hee Won Lee

36 Liberty Ridge Rd, Basking Ridge, NJ 07920, USA
knowpd@gmail.com • +1 (919) 800-8993

SUMMARY

Cloud/AI infrastructure and software engineer with deep expertise in designing and optimizing distributed systems and developing LLM-powered multi-agent systems. At Samsung, led the Cloud Platform team in building and operating cloud infrastructure services across cloud and on-premises datacenters. Developed Kubernetes platforms, Ceph Storage, GitOps pipelines, observability stacks, infrastructure as code, and GenAI services. At AT&T, developed Kubernetes-orchestrated Ceph storage and Redis in-memory databases. Also, designed and implemented network systems for OpenStack clusters. Holds a Ph.D. in Computer Science with a focus on virtualization (QEMU-KVM), software-defined networking (SDN), and systems performance optimization.

EXPERIENCE

Samsung Electronics, Hwaseong, South Korea

- Principal Engineer — Dec 2020 – Feb 2024
  - Hands-on Software Development (Kubernetes, Ceph, GitOps, Observability, IaC, and GenAI)
    - Developed a highly-available cluster service using open-source Kubernetes by deploying control plane components (kube-apiserver, etcd, etc) behind a load balancer in a dedicated management cluster
    - Developed zero-touch provisioning for Kubernetes and Ceph storage clusters on Azure VM Scale Sets (w/ Rook)
    - Developed K8s Custom Resources and Controller for HPC batch scheduler using the Operator SDK (w/ Golang)
    - Developed GitHub Actions pipelines (w/ SonarQube) to deploy apps to Kubernetes clusters (w/ Argo CD)
    - Developed observability stacks for K8s services w/ Jaeger, Prometheus, and Grafana using OpenTelemetry (w/ gRPC)
    - Developed telemetry systems for VM services with Telegraf, InfluxDB, and Grafana
    - Developed infrastructure as code for Azure and AWS using ARM template/Bicep and Terraform
    - Developed an LLM-powered multi-agent chat service using retrieval-augmented generation (RAG) (w/ LangChain and LangGraph)
  - Technical Leadership
    - As Head of Group for Cloud Platform (reporting to the SVP), led eight projects, including:
      1. Software development – HA Kubernetes service, HPC batch scheduler, and cloud cost monitoring system;
      2. System engineering – VMware VM/VDI services, OpenStack VM service, and HA Active Directory;
      3. Public cloud architecture – Azure/AWS governance framework (landing zone) and AWS Outposts evaluation.
    - Delivered large-scale infrastructure services – VM (5K+ instances), VDI (10K+ instances), HPC batch scheduler (10K+ servers), HA Kubernetes (500+ servers), and HA Active Directory (60K+ users)
    - Managed 24/7 VM/VDI services with a 99.95% SLO (< 4 hours downtime/year) / zero downtime for Acitve Directory
    - Led team growth from 30 to over 60 engineers (SW/DevOps Engineers and Cloud Architects) within three years
    - Defined MBOs and developed KPIs for team projects and achieved 100% goal completion
    - Developed security best practices for Azure with Zero Trust (MFA, JIT, and Sentinel) and for Kubernetes (image scan w/ Trivy, container runtime w/ non-root user, secret w/ Azure Key Valut, and ingress w/ TLS/mTLS)

AT&T Labs Research, Bedminster, New Jersey, USA

- Principal Inventive Scientist — Mar 2015 – Jul 2020
  - Kubernetes & Observability
    - Designed and implemented containerized Ceph with Kubernetes helm charts (with an one-osd-per-pod approach)
    - Developed real-time monitoring systems for containerized Ceph with Telegraf, InfluxDB, and Grafana
    - Designed and implemented Redis High Availability with Kubernetes in CI/CD environment (Gerrit and Jenkins)
    - Integrated Redis HA helm charts into Radio Access Network Intelligent Controller release 4 (field trial version)
  - Storage Systems
    - Designed and implemented I/O traffic protection mechanisms for cloud storage
    - Designed and implemented an iSCSI block storage backend with a commodity SSDs for OpenStack clusters
    - Designed and implemented the backend of OpenStack Cinder for dynamic configuration and QoS support
    - Evaluated cache performance across host and VM page cache, dm-cache, and Ceph RBD cache
    - Evaluated the latency of Redis with Intel Optane Persistent Memory using the *memtier_benchmark* tool
    - Evaluated and compared the performance of *Redis* vs *Aerospike* vs *Cassandra* using the YCSB benchmark tool
    - Designed and implemented high-performance VMs by applying CPU pinning and SR-IOV PCI passthrough
  - Network Systems
    - Designed and implemented a system that dynamically controls network bandwidth and latency between OpenStack Neutron deployments with Ansible (using Open vSwitch and Tc/NetEm)
    - Set up a testbed for RoCE-based lossless networks (using PFC-capable Mellanox Spectrum Switch and ConnectX-5 NICs) and evaluated the performance of NVMe over Fabric for zero CPU usage on the target host machine
    - Evaluating the performance of Apache Traffic Server with Intel Optane Persistent Memory
    - Analyzed cost benefits by applying Intel Optane Persistent Memory to AT&T's Content Delivery Network (CDN)

North Carolina State University, Raleigh, North Carolina, USA

- Research Assistant — Jan 2010 – Dec 2014
  - Designed and implemented a virtualization platform for evaluating distributed applications running on diverse OSs (Linux, FreeBSD, Windows, JunOS)
  - Redesigned and customized the QEMU-KVM hypervisor to synchronize distributed VMs with virtual time

Korea Telecom, Daejeon, South Korea

- Software Engineer — Jan 2002 – Jul 2009
  - Capacity Planning for KT Backbone Network and IPTV Services
    - Built a virtual network environment of KT backbone networks using OPNET simulation tool
    - Performed link capacity planning for IPTV services
  - VoIP Network Management System
    - Conducted requirement management and then designed architecture of VoIP-NMS
    - Deployed VoIP-NMS to production
  - IP Core Network Management System
    - Designed and implemented a system component that collects network equipment information from MIB (Management Information Base) using SNMP (Small Network Management Protocol)
    - Developed SQL scripts for Oracle database
    - Developed a system component that exchanges information with Service Assurance System using XML-RPC
    - Set up an MPLS testbed using Cisco Routers 7204/7500, Juniper Router M5 and Ixia Traffic Generator

CERTIFICATIONS

Microsoft Azure Solutions Architect Expert (AZ-305) / Azure Administrator Associate (AZ-104)

Microsoft Cybersecurity Architect Expert (SC-100) / Azure Security Engineer Associate (AZ-500)

EDUCATION

North Carolina State University, Raleigh, North Carolina, USA

- Ph.D. in Computer Science — Aug 2009 – May 2015
  - Concentration: Virtualization, Networked Systems, Distributed Systems

Carnegie Mellon University, Pittsburgh, Pennsylvania, USA

- M.E. in Software Engineering — Aug 2004 – Aug 2005

SOFTWARE DEVELOPMENT EXPERIENCE

CPU & Memory

- CPU/Memory Control [Python]
  - Designed and implemented a CPU/Memory control algorithm that enforces CPU/Memory ceiling and allocates a relative share of CPU time using Linux Control Groups (Cgroups)
- Multi-Threading [C++]
  - Created a thread to search text file paths and multiple worker threads to count words using Boost Filesystem and Thread libraries.
  - Implemented multiple producer, multiple consumer thread-safe queue and map
- Prioritized Preemptive Schedulers [C]
  - Implemented a prioritized preemptive scheduler using POSIX thread library
  - Designed and implemented a priority-based scheduling algorithm using XINU kernel (a small UNIX OS)
- Demand Paging [C]
  - Implemented a demand paging system that allows for more address space than physically available one (XINU)

Networking & IO

- I/O Traffic Protection [Python]
  - Designed and implemented an I/O bandwidth reservation algorithm that guarantees the minimum I/O bandwidth of a virtual host connected to a shared storage volume through iSCSI
  - Designed a NUMA-aware CPU pinning algorithm that protects I/O traffic from CPU interference
  - Evaluated the system performance for diverse types of I/O pattern using FIO and Filebench
- Network Link Emulation [C++, Python]
  - Interconnected distributed VMs through virtual links using Tc/NetEm with adaptive time dilation
- Software-Defined Networking [Python]
  - Used OpenFlow POX controller and Open vSwitch to migrate VirtualBox-based virtual networks
- Routing Algorithms [Java]
  - Implemented modified Dijkstra’s algorithm with negative weights
  - Implemented vertex/edge disjoint paths and elementary circuits search algorithms
  - Implemented CPLEX code with Integer Linear Programming for Routing and Wavelength Assignment

- Queueing Systems [Java]
  - Implemented discrete event simulation models for queuing systems (M/M/1, M/M/m, M/G/m)
- Wireless Networking [C++]
  - Designed and implemented an OMNeT++ model for IVG (Inter-Vehicle Geocast) routing protocol in vehicle-to-vehicle ad-hoc network simulation
  - Connected KVM-based virtual nodes through a wireless simulation model of OMNeT++
- Peer-to-Peer File Sharing System [Java, Apache Mina, Apache FtpServer, JavaDB]
  - Mapped a file signature (created by encoding text document with m-bit string) onto a hash space

Virtualization

- Hypervisor (QEMU-KVM) [C++]
  - Designed and implemented a spinlock on shared memory to synchronize distributed VMs (processes created by QEMU-KVM hypervisor) and simulation nodes (processes created by NS-3) with virtual time
  - Implemented a virtual time synchronization daemon with TCP/UDP sockets using Boost Asio Library
- High-Performance Emulation System [C++, Python]
  - Emulated higher performance with a time dilation technique using unmodified OSs (Linux, FreeBSD, Windows, and Junos) and real application workloads (VLC media player)
  - Modified the QEMU-KVM hypervisor for virtual time
- Hybrid Simulation and Emulation System [C++, Python]
  - Built an integrated simulation (NS-3) and emulation (QEMU-KVM) framework in a distributed environment

PUBLICATIONS

DISTRIBUTED STORAGE SYSTEMS

- **MIST: Mitigating Host-Side Interference for Storage Traffic in Virtualized Data Centers**.
  Hee Won Lee and Moo-ryong Ra
  In *IEEE 9th International Conference on Cloud Computing (CLOUD 2016)*.
- **Fighting with Unknowns: Estimating the Performance of Scalable Distributed Storage Systems with Minimal Measurement Data**.
  Moo-ryong Ra and Hee Won Lee.
  In *IEEE 35th Symposium on Mass Storage Systems and Technologies (MSST 2019)*.
- **eMRC: Efficient Miss Ratio Approximation for Multi-Tier Caching**.
  Zhang Liu, Hee Won Lee, Yu Xiang, Dirk Grunwald, and Sangtae Ha.
  In *USENIX 19th Conference on File and Storage Technologies (FAST 2021)*.

NETWORKED SYSTEMS

- **High-performance Emulation of Heterogeneous Systems using Adaptive Time Dilation**.
  Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
  International Journal of High Performance Computing Applications, vol. 29, issue 2, May 2015.
- **Integrated Simulation and Emulation using Adaptive Time Dilation**.
  Hee Won Lee, David Thuente, and Mihail L. Sichitiu.
  In *ACM 2nd SIGSIM Conference on Principles of Advanced Discrete Simulation (PADS 2014)*.
- **Network Link Emulation With Adaptive Time Dilation**.
  Hee Won Lee, Mihail L. Sichitiu, and David Thuente.
  Journal of Parallel and Distributed Computing, vol. 104, pp. 88–98, Jun 2017.
- **Accelerating Applications in the Fast-Moving Devices with Proactive Provisioning (Poster)**.
  HyunJong Lee, Hee Won Lee, Moo-Ryong Ra, Yu Xiang, and Jason Flinn.
  In *Proceedings of the 17th Annual International Conference on Mobile Systems, Applications, and Services (MobiSys 2019)*.

PATENTS

Dynamic Provisioning of Storage in the Cloud.
Moo-ryong Ra and Hee Won Lee.
US 10,530,703 B2 – Date of Patent: January 7, 2020.

[*CV compiled on 2025-06-30*]